SIMULATION EXPERIMENT – 6

Simulation-Based Investigation of Microwave Semiconductor Devices and Frequency-Conversion Characteristics of a Microwave Mixer

Objectives of this Experiment:

1. To develop simulation models representing the operating characteristics of selected microwave semiconductor devices.

A. Mixer Frequency Conversion

Aim

To simulate frequency conversion in a microwave mixer using an RF input frequency of 10 GHz and a local oscillator frequency of 9.5 GHz, with RF input power of −10 dBm, LO power of +7 dBm, and a 50 Ω intermediate-frequency load. To obtain the output spectrum, identify the intermediate and sum frequencies, and calculate the percentage deviation between theoretical and simulated frequencies.

Software used

1. MATLAB

CODE-

```
clc;
clear;
close all;

% EXPERIMENT 6(A): Mixer Frequency Conversion

% Input parameters
fRF = 10;          % RF frequency in GHz
fLO = 9.5;         % LO frequency in GHz
PRF = -10;         % RF input power in dBm
PLO = 7;           % LO input power in dBm
ZIF = 50;          % IF load in ohms

% Theoretical mixing frequencies
```

```
fIF = abs(fRF-fLO);
fSUM = fRF+fLO;

fprintf('MICROWAVE MIXER FREQUENCY CONVERSION\n');
fprintf('RF frequency = %.3f GHz\n',fRF);
fprintf('LO frequency = %.3f GHz\n',fLO);
fprintf('RF input power = %.2f dBm\n',PRF);
fprintf('LO input power = %.2f dBm\n',PLO);
fprintf('IF load = %.0f ohms\n',ZIF);
fprintf('Theoretical IF frequency = %.3f GHz\n',fIF);
fprintf('Theoretical sum frequency = %.3f GHz\n',fSUM);

% Illustrative spectrum
% Relative component amplitudes are assumed, not simulated
freq = [fIF, fRF, fLO, fSUM];
power = [-17, PRF, PLO, -25];

figure;
stem(freq,power,'filled','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Illustrative Power (dBm)');
title('Microwave Mixer Output Spectrum');
xticks([0.5 9.5 10 19.5]);
xlim([0 21]);

% Simulated frequency values should be replaced by actual values
fIF_sim = 0.5;       % Replace with measured/simulated IF (GHz)
fSUM_sim = 19.5;     % Replace with measured/simulated sum (GHz)
```

devIF = abs(fIF_sim-fIF)/fIF*100;

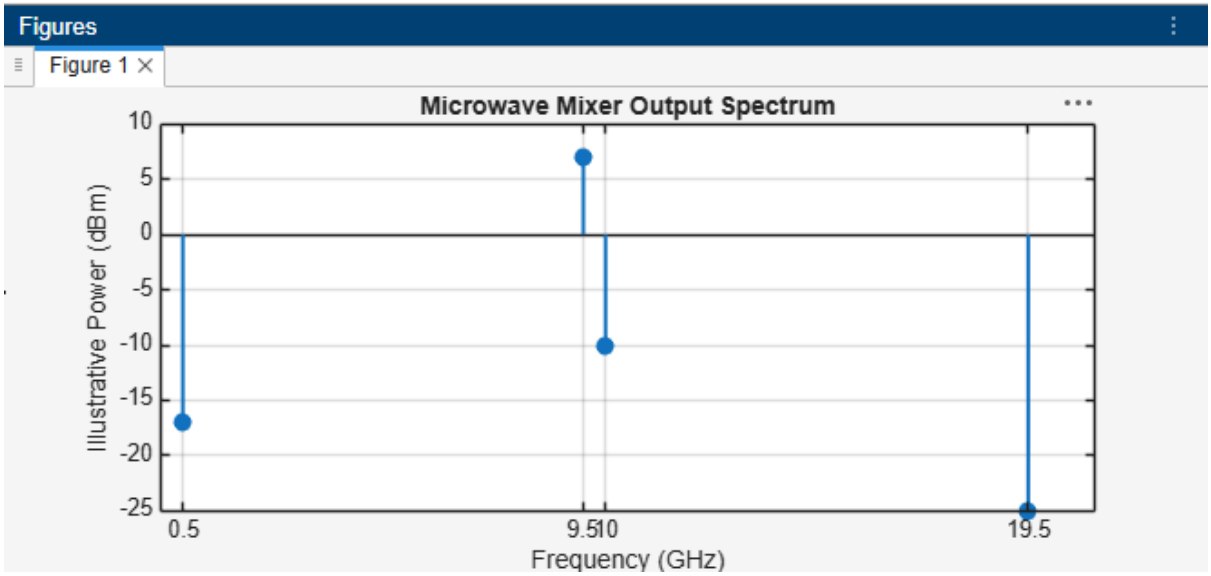
devSUM = abs(fSUM_sim-fSUM)/fSUM*100;

fprintf('\nFrequency deviation:\n');

fprintf('IF deviation = %.3f %%\n',devIF);

fprintf('Sum-frequency deviation = %.3f %%\n',devSUM);

OUTPUT-



Result

The theoretical intermediate frequency was calculated as 0.5 GHz, and the sum frequency was calculated as 19.5 GHz. An illustrative mixer output spectrum was plotted using MATLAB, and the percentage frequency deviations were calculated by comparing the entered simulated frequencies with the theoretical values. Actual spectrum amplitudes and generated components must be verified using a nonlinear mixer simulation.

B) Conversion Loss: Using the same mixer, record the IF output power at 0.5 GHz. Calculate conversion loss CL = PRF − PIF and compare it with the expected passive-mixer range of approximately 6–10 dB. Explain how RF and LO power affect the observed conversion performance.

Aim

To determine the conversion loss of the microwave mixer using the RF input frequency of 10 GHz and LO frequency of 9.5 GHz by recording the IF output power at 0.5 GHz. To calculate the conversion loss and compare it with the expected passive-mixer range of approximately 6–10 dB, and analyze the effects of RF and LO input powers on conversion performance.

Software used

1. MATLAB

CODE-

```
clc;
clear;
close all;

% EXPERIMENT 6(B): Conversion Loss

% Mixer input parameters
fRF = 10;            % RF frequency (GHz)
fLO = 9.5;           % LO frequency (GHz)
fIF = abs(fRF-fLO);  % IF frequency (GHz)

PRF = -10;           % RF input power (dBm)
PLO = 7;             % LO input power (dBm)

% Assumed IF output power for demonstration
% Replace with actual simulated/measured value
PIF = -17;           % IF output power (dBm)

% Conversion loss
CL = PRF-PIF;

fprintf('MICROWAVE MIXER CONVERSION LOSS\n');
fprintf('RF frequency = %.2f GHz\n',fRF);
fprintf('LO frequency = %.2f GHz\n',fLO);
fprintf('IF frequency = %.2f GHz\n',fIF);
fprintf('RF input power = %.2f dBm\n',PRF);
fprintf('LO input power = %.2f dBm\n',PLO);
```

```
fprintf('IF output power = %.2f dBm\n',PIF);
fprintf('Conversion loss = %.2f dB\n',CL);

% Compare with expected passive mixer range
if CL >= 6 && CL <= 10
   fprintf('Conversion loss is within the expected 6-10 dB range.\n');
elseif CL < 6
   fprintf('Conversion loss is below the expected range.\n');
else
   fprintf('Conversion loss is above the expected range.\n');
end

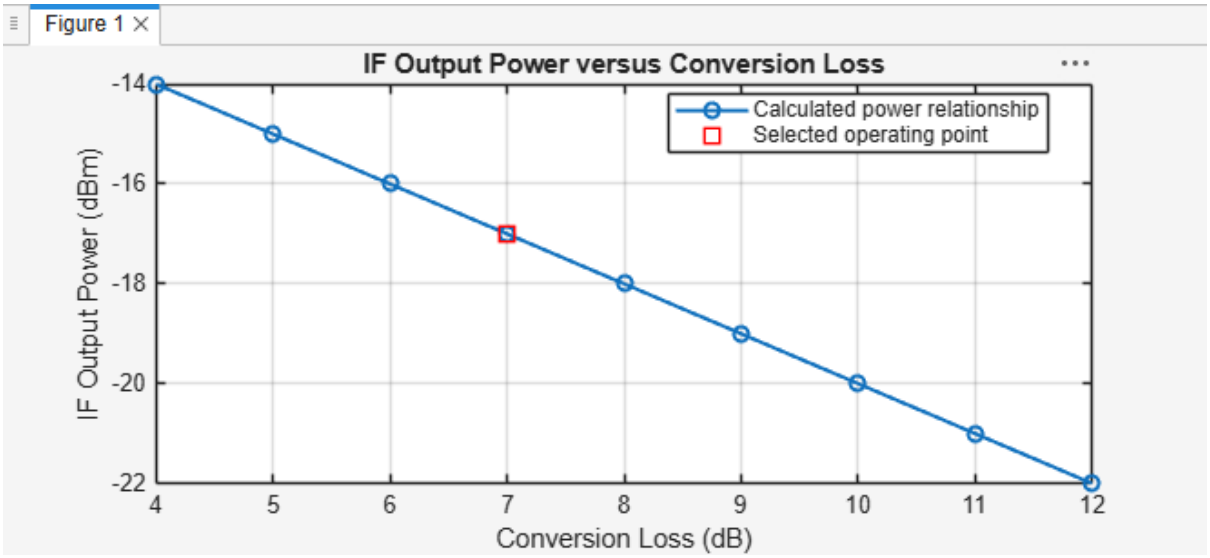
% Illustrative IF output power versus conversion loss
CL_values = 4:1:12;
PIF_values = PRF-CL_values;

figure;
plot(CL_values,PIF_values,'o-','LineWidth',1.5);
hold on;
plot(CL,PIF,'rs','MarkerSize',8,'LineWidth',1.5);
grid on;
xlabel('Conversion Loss (dB)');
ylabel('IF Output Power (dBm)');
title('IF Output Power versus Conversion Loss');
legend('Calculated power relationship','Selected operating point',...
      'Location','best');
```

OUTPUT-

Command Window

```
MICROWAVE MIXER CONVERSION LOSS
RF frequency = 10.00 GHz
LO frequency = 9.50 GHz
IF frequency = 0.50 GHz
RF input power = -10.00 dBm
LO input power = 7.00 dBm
IF output power = -17.00 dBm
Conversion loss = 7.00 dB
Conversion loss is within the expected 6-10 dB range.
>>
```



Result

The mixer conversion loss was calculated using the RF input power and IF output power at 0.5 GHz.